

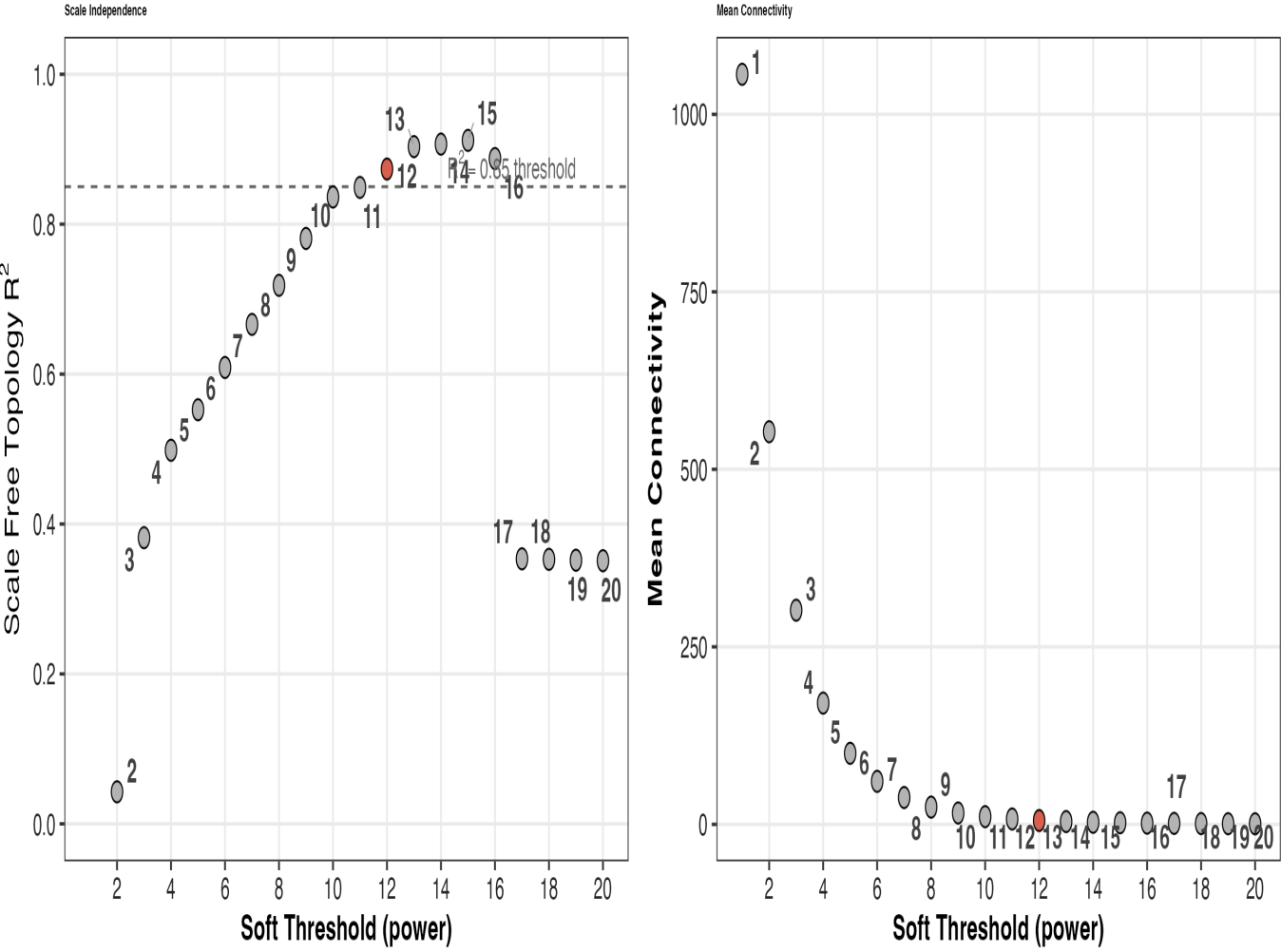
S5a – WGCNA Quality Control

A: Soft threshold | B: Dendrogram | C: Compartment enrichment | D: Module–trait heatmap | E: Bicor sensitivity

A

Scale-Free Topology Fit

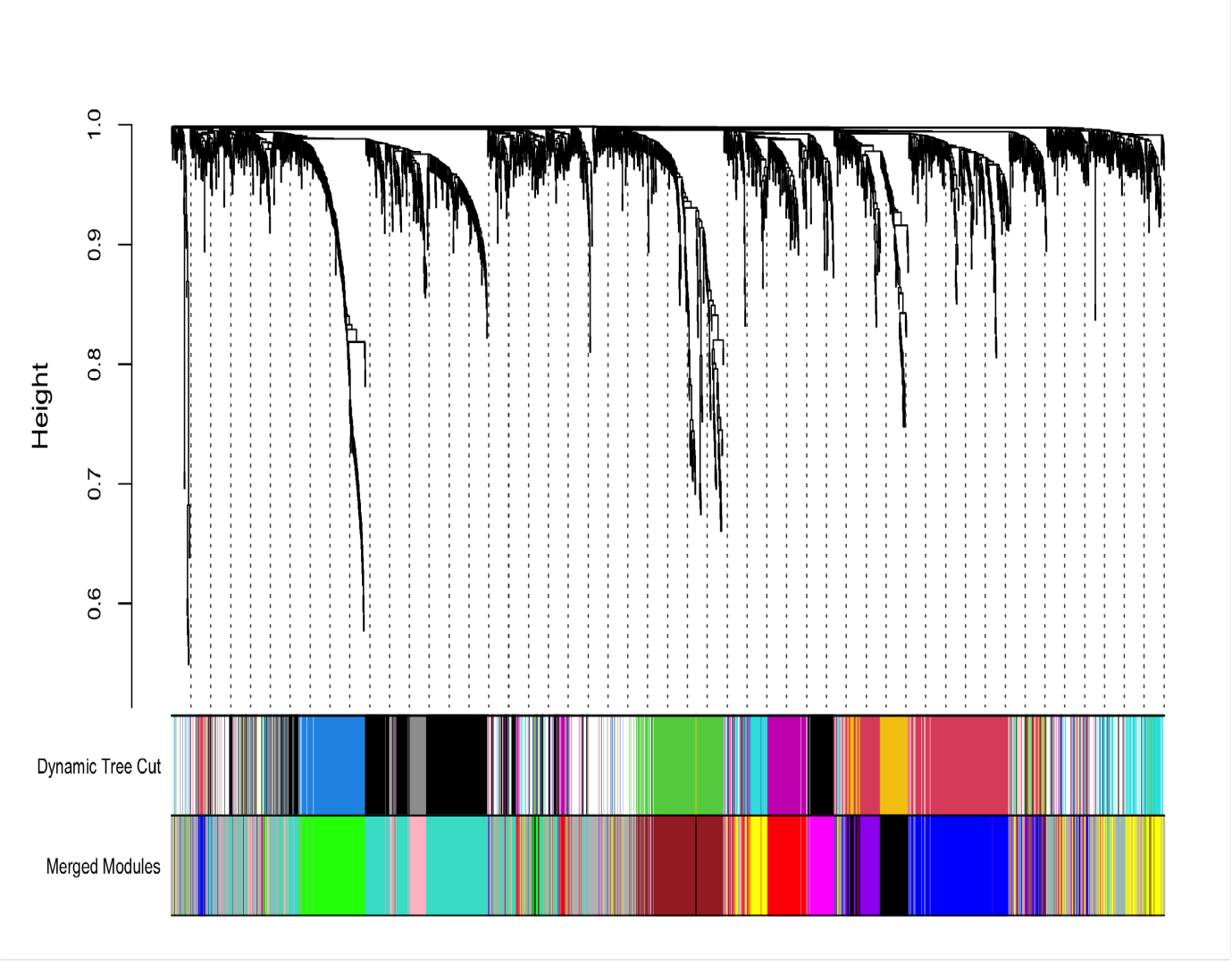
Signed network | selected power = 12 ($R^2 = 0.874$) | 2,113 proteins



B

Protein Dendrogram & Module Colors

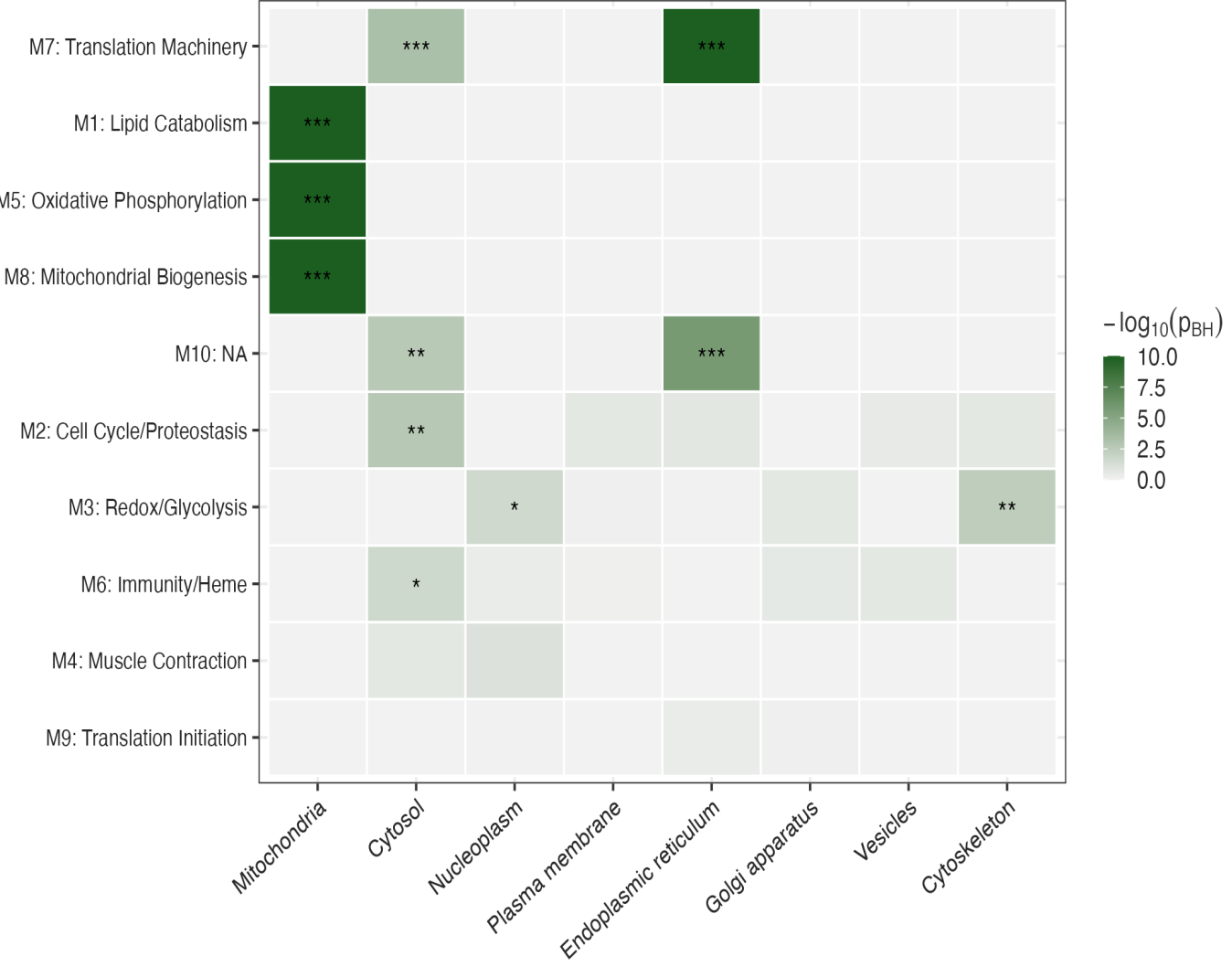
Signed network | power = 12 | 10 modules | 2,113 proteins (507 unassigned)



C

Module Subcellular Compartment Enrichment

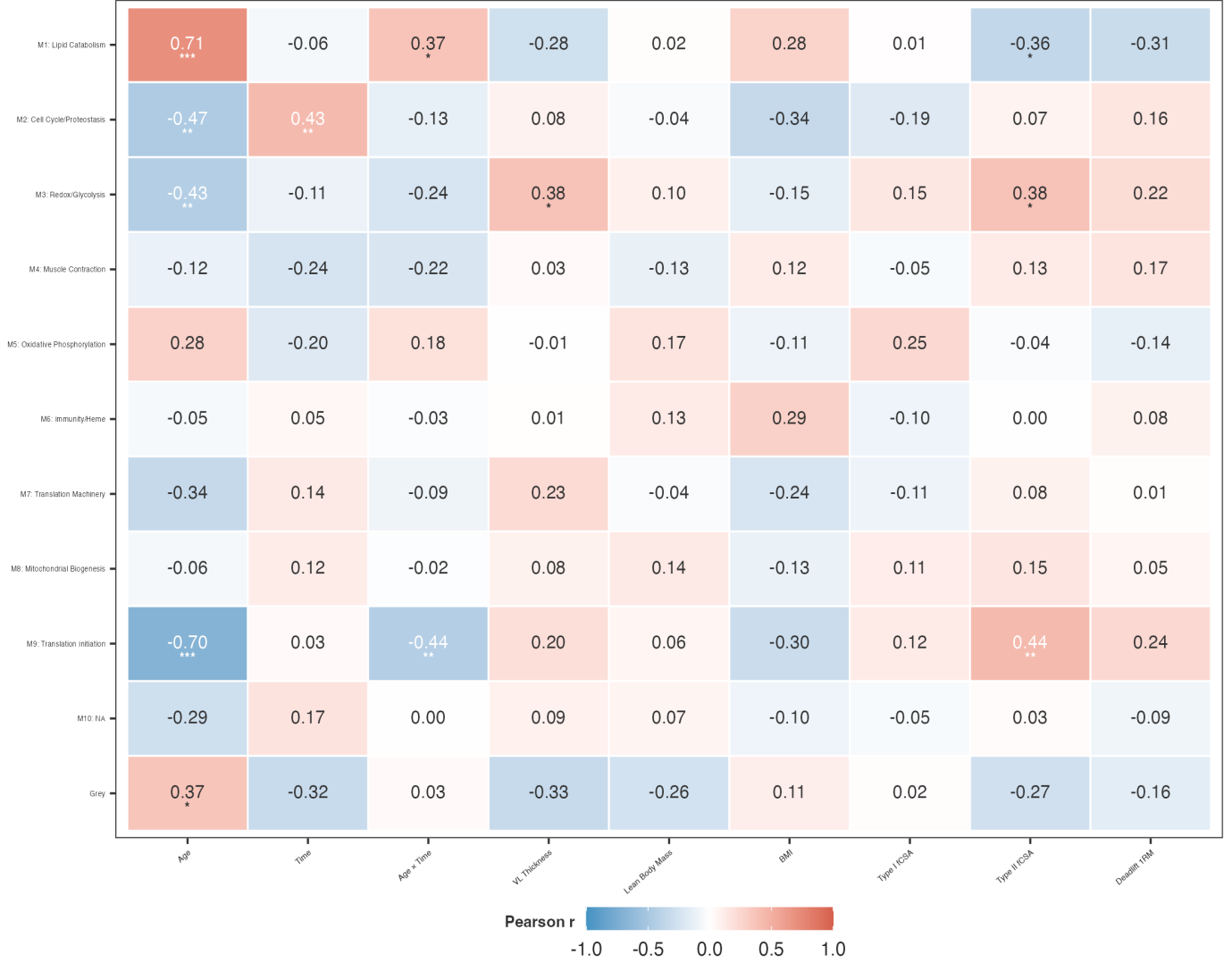
One-sided Fisher exact test, BH-corrected (80 tests) | * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$



D

Module–Trait Correlations

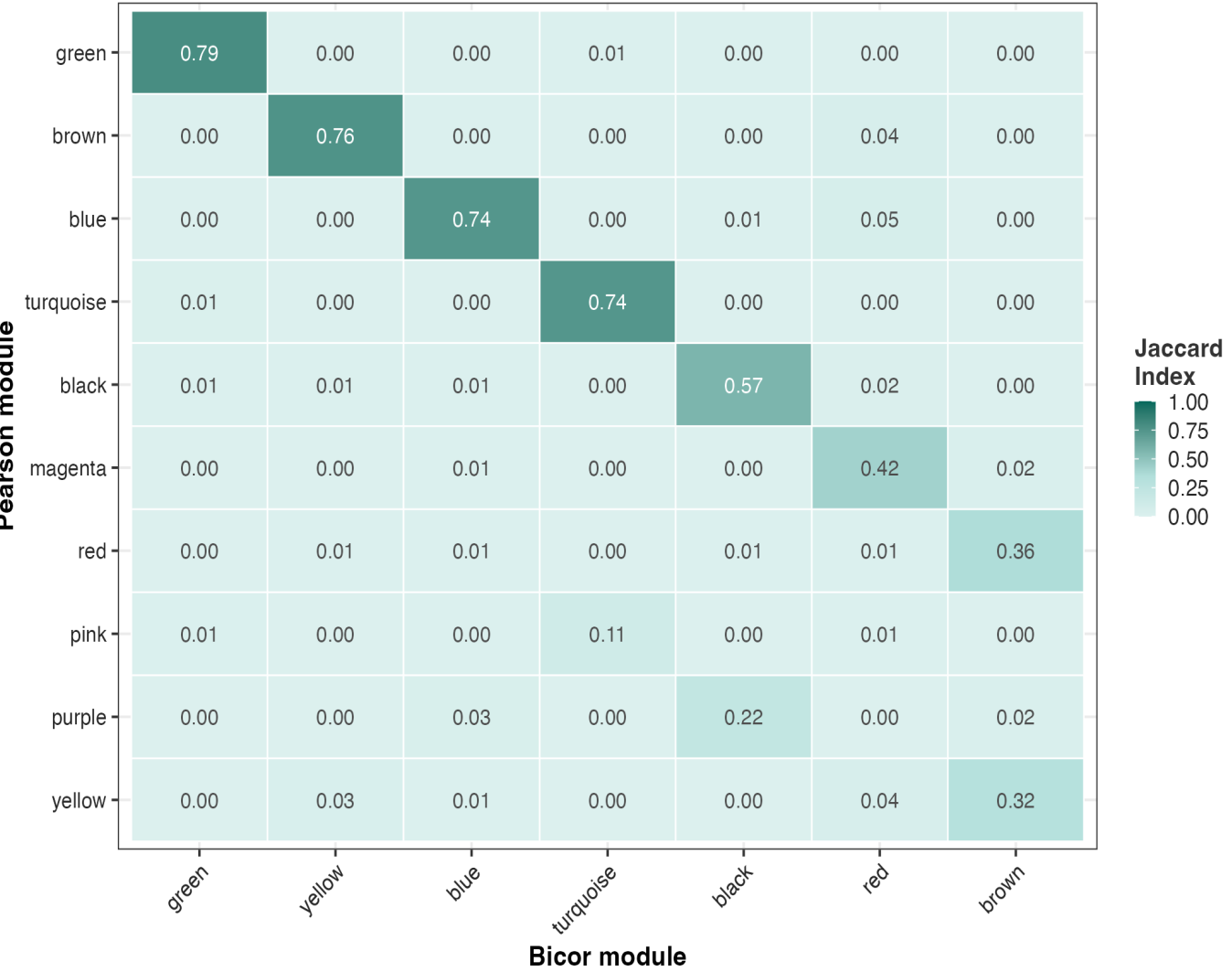
Pearson correlation | * $BH < 0.05$ ** $BH < 0.01$ *** $BH < 0.001$



E

Pearson vs Bicor Module Overlap

Mean matched Jaccard = 0.628 | 5/7 modules > 0.5



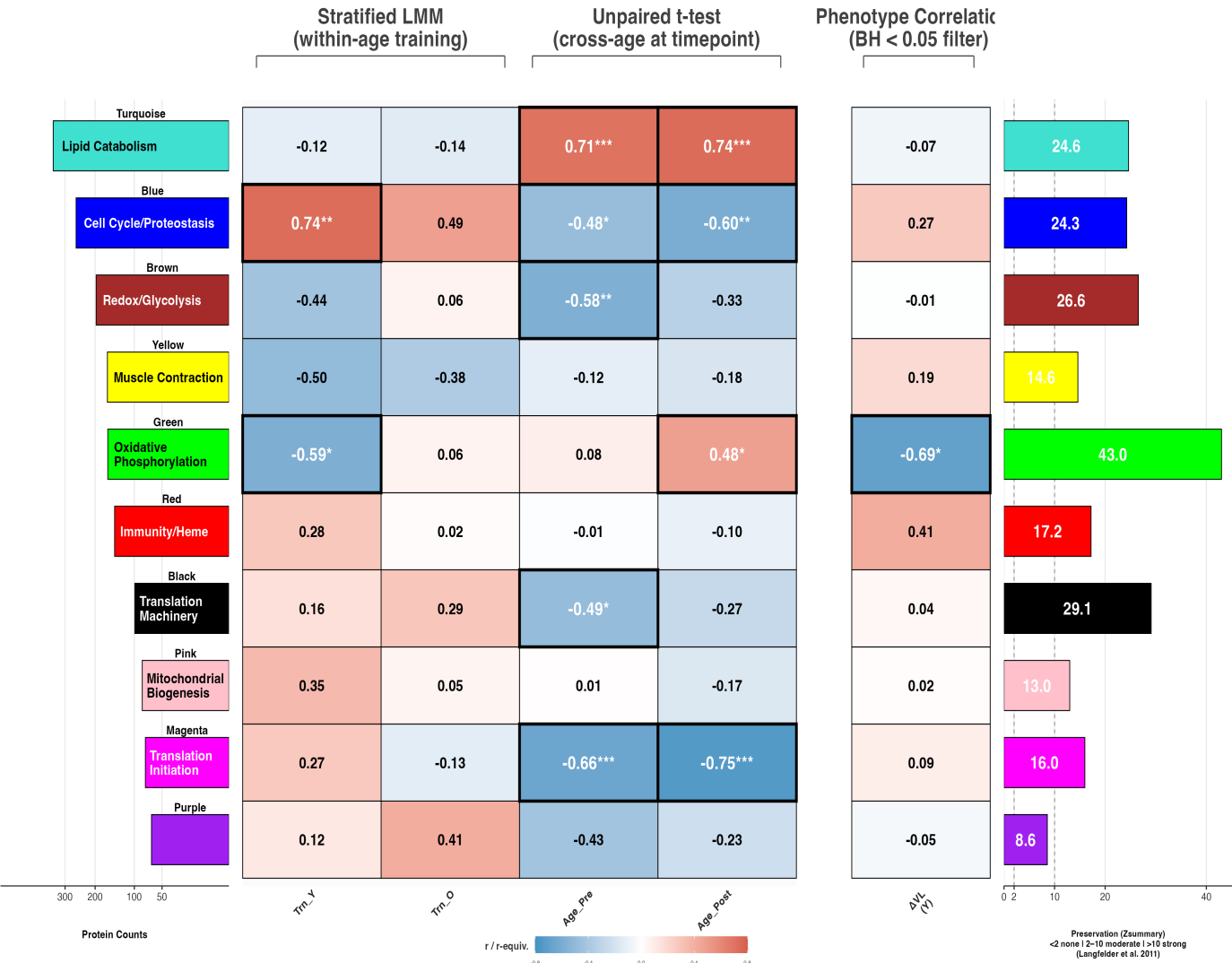
S5b – Module–Trait Analysis

A: Age–stratified LMM | B: Exemplar strip | C: DEP–module UpSet | D: ME trajectories | E: Preservation vs effect

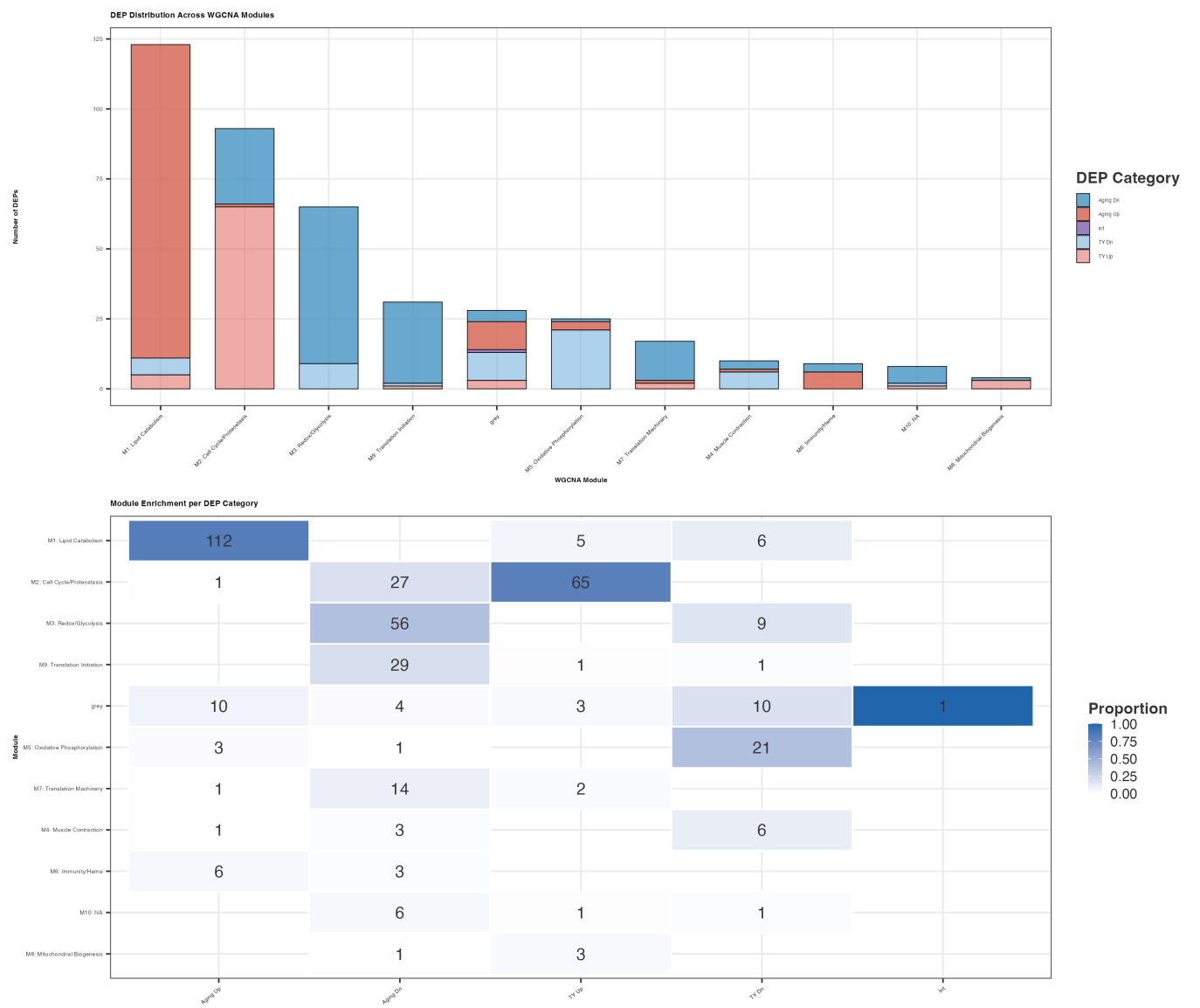
A

Age-Stratified Analysis — Training, Aging, & Phenotype Associations

10 modules | LMM: eigengene ~ time + (1|subject), BH per stratum | Aging: unpaired t-test, BH per timepoint | Phenotype: Pearson r, BH per column
* p<0.05 ** p<0.01 *** p<0.001 † p<0.1 | Black border = BH < 0.05



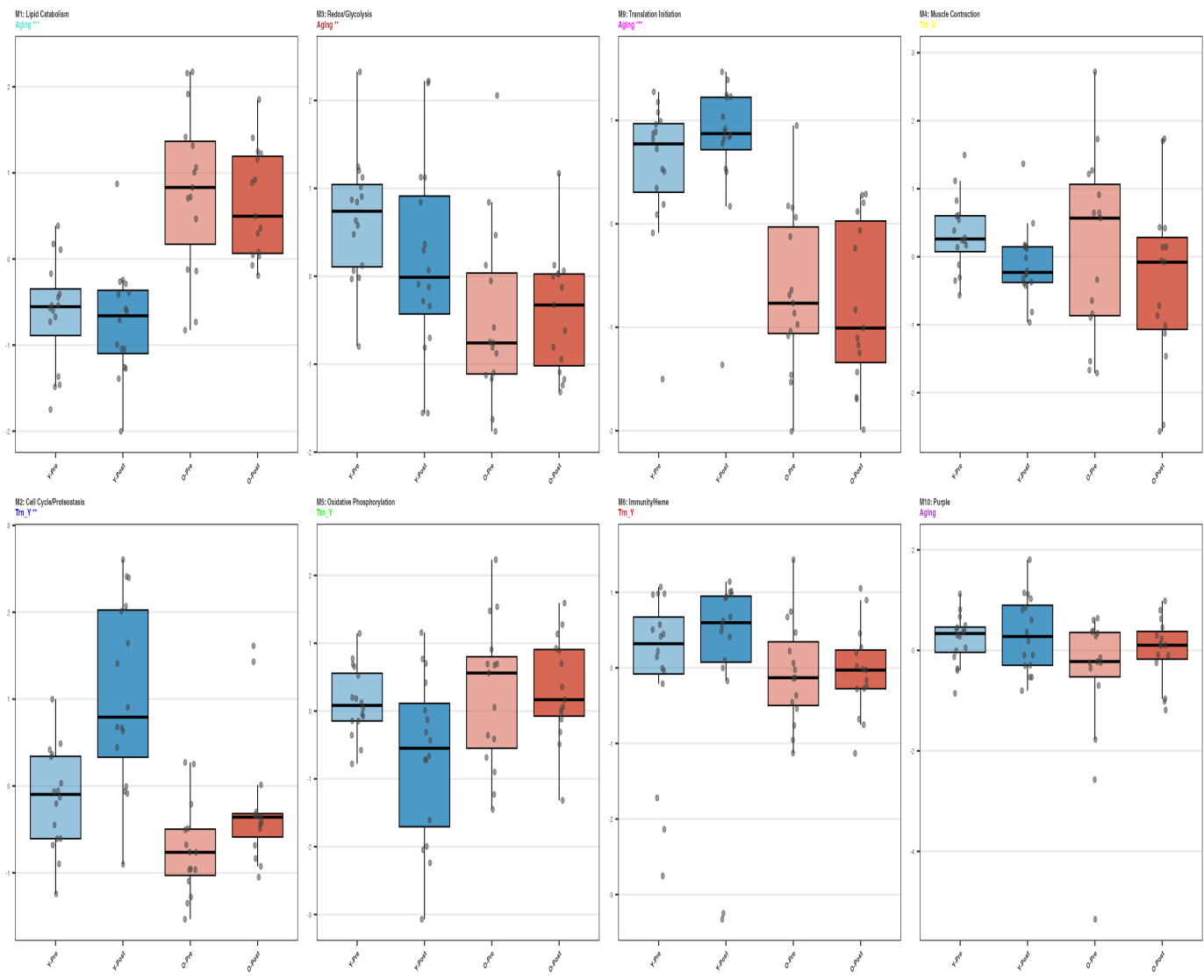
C



B

Eigengene Exemplars – 8 modules (BH<0.05 or 10<0.4)

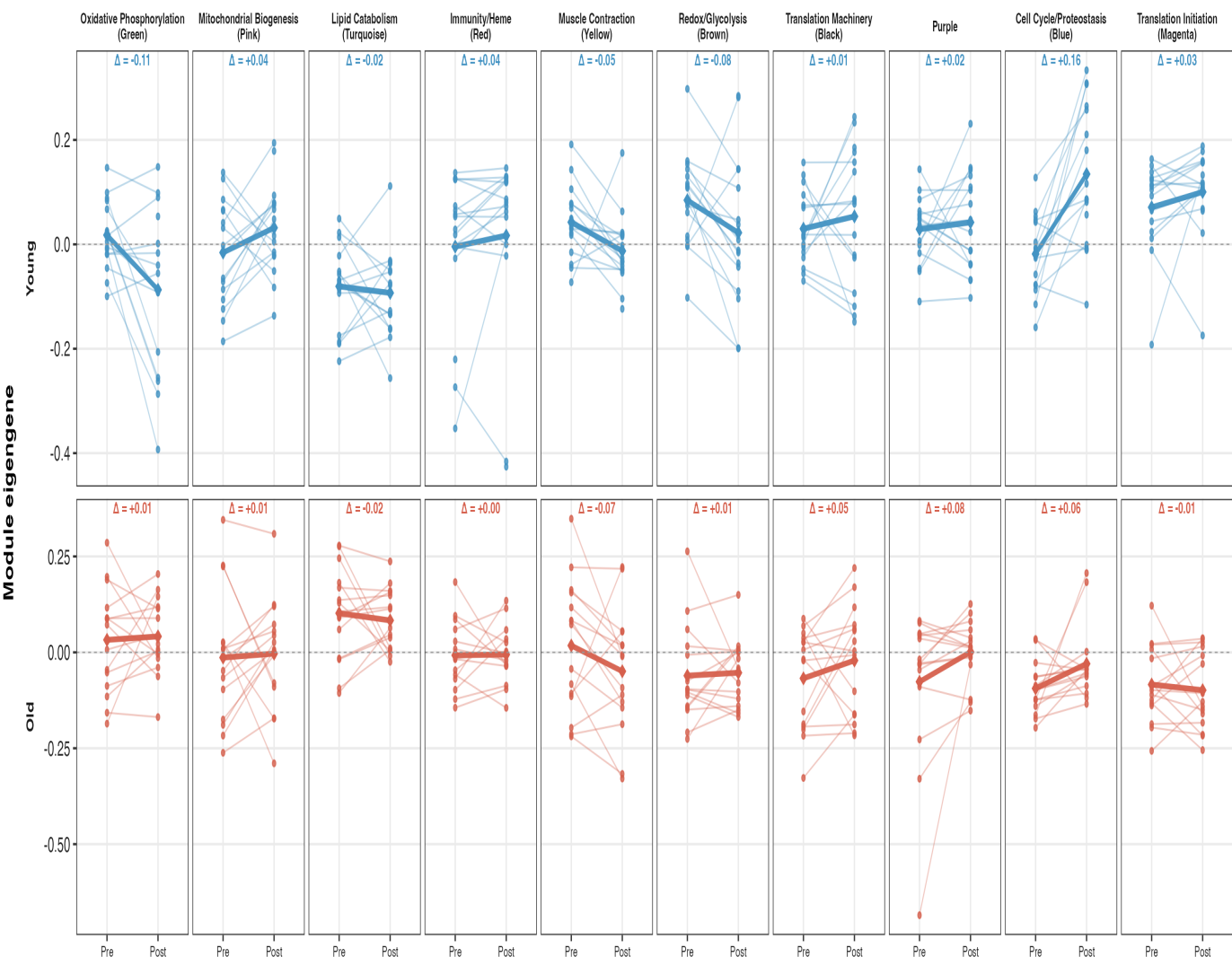
Z-scored module eigengenes by Group x Time | Aging: *** negative | Trn_O yellow | Trn_Y *** blue | Trn_Y, Trn_O green | Trn_Y, Trn_O red | Aging: purple



D

Module Eigengene Trajectories (Pre → Post)

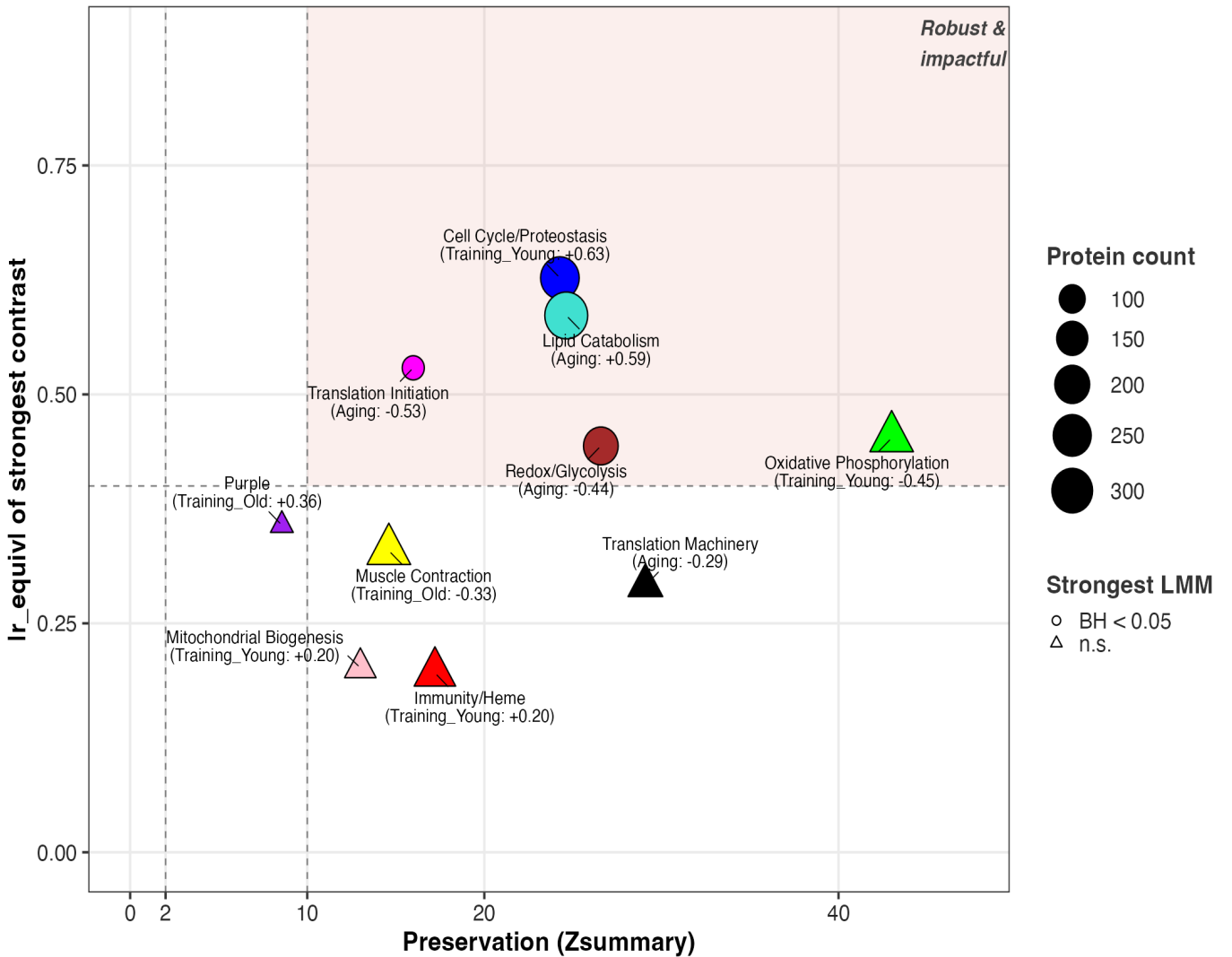
Per-subject ME paths | bold line = group mean | diamonds = group means



E

Module Preservation vs. Strongest Effect Size

Zsummary (Langfelder 2011) | Ir_equiv from LMM contrast audit



S5c – Module Triptychs (Key Modules)

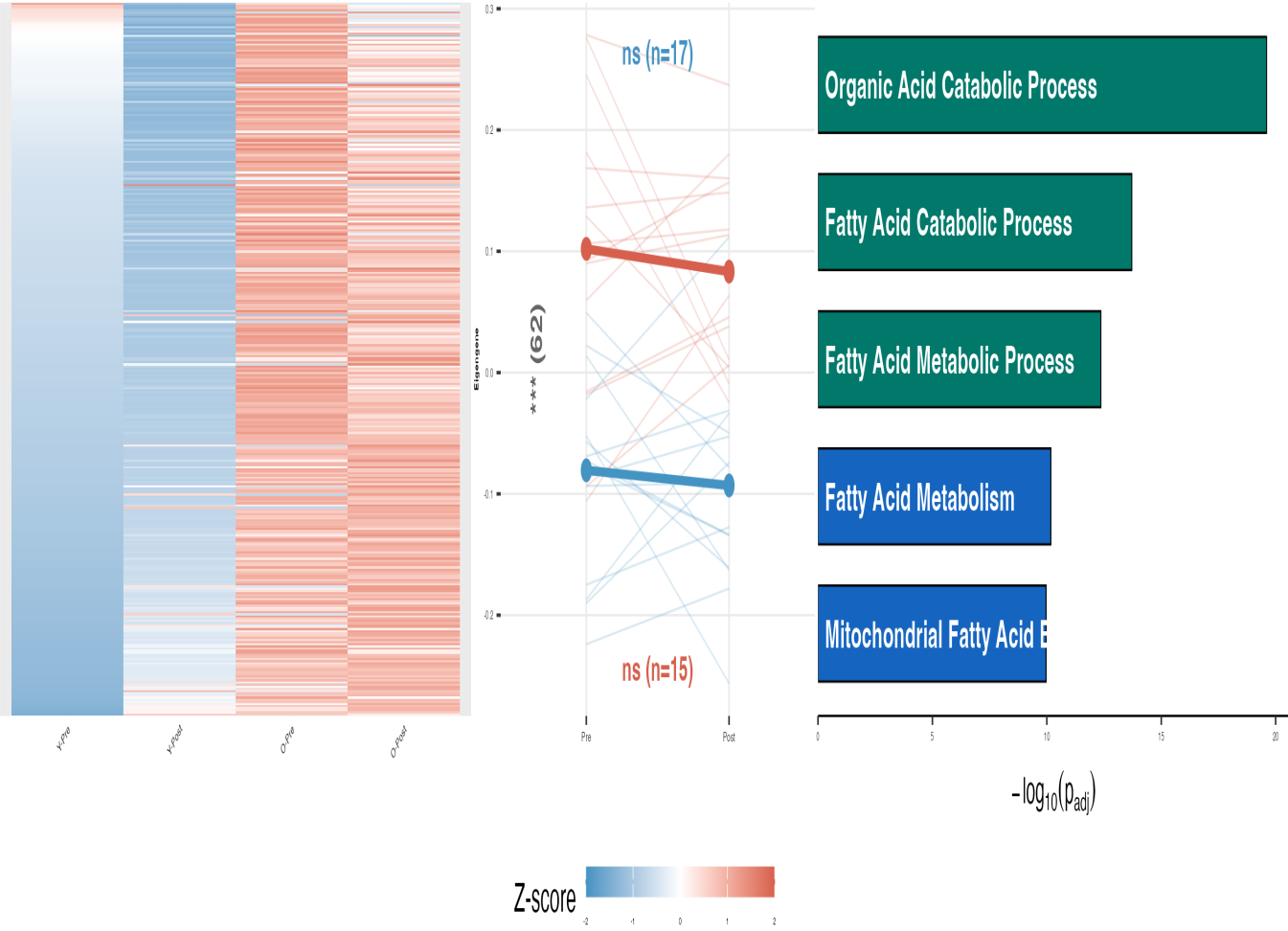
A: Turquoise (Lipid) | B: Blue (Cell Cycle) | C: Green (OxPhos) | D: Brown (Redox) | E: Magenta (Translation)

A

M1: Lipid Catabolism

Aging: $r = +0.59$

M1: Lipid Catabolism (n=334)



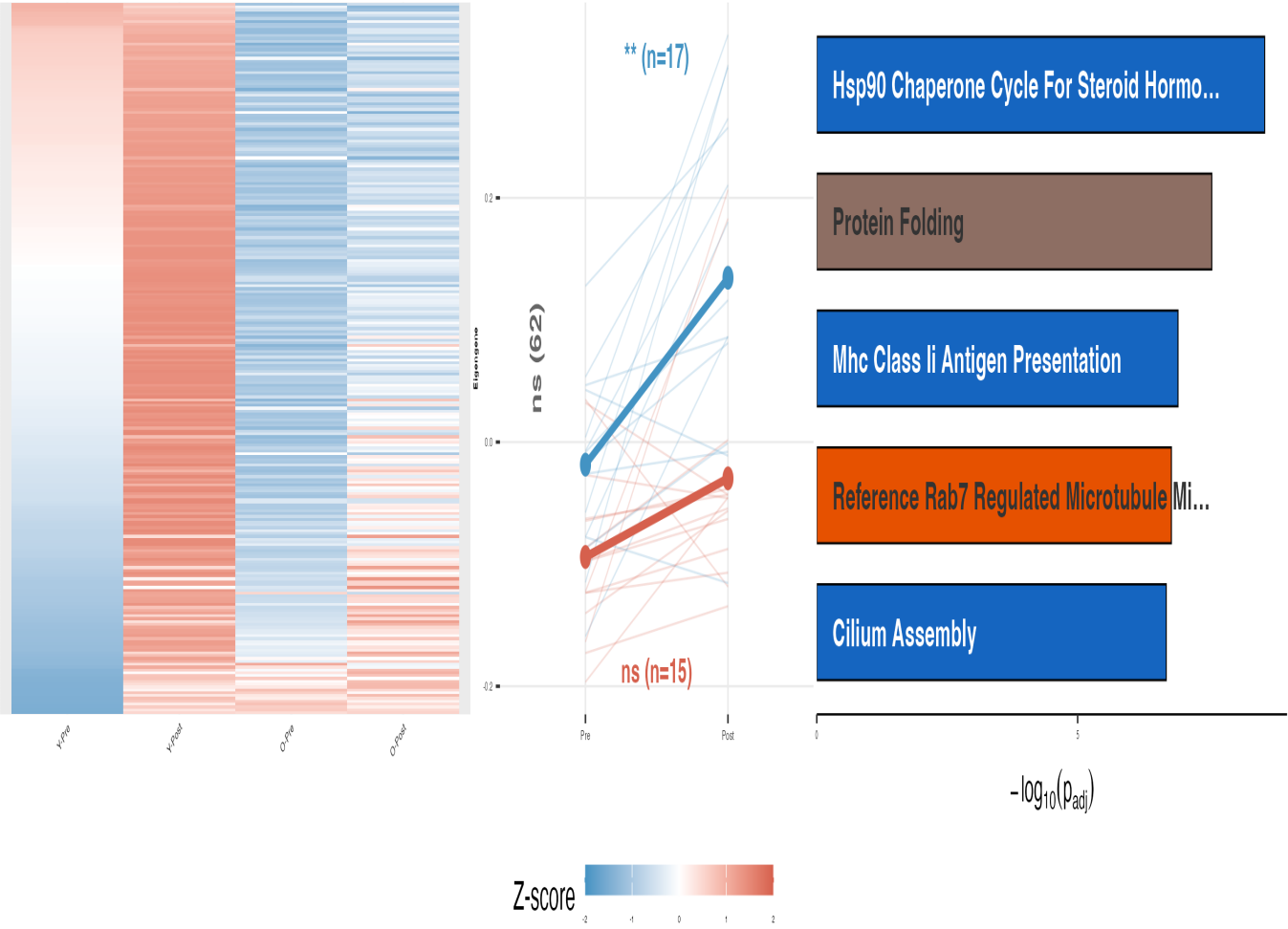
*FDR < 0.5 **FDR < 0.1 ***FDR < 0.01 (all LMM Bonferroni)

B

M2: Cell Cycle/Proteostasis

Training Young: $r = +0.62$

M2: Cell Cycle/Proteostasis (n=250)



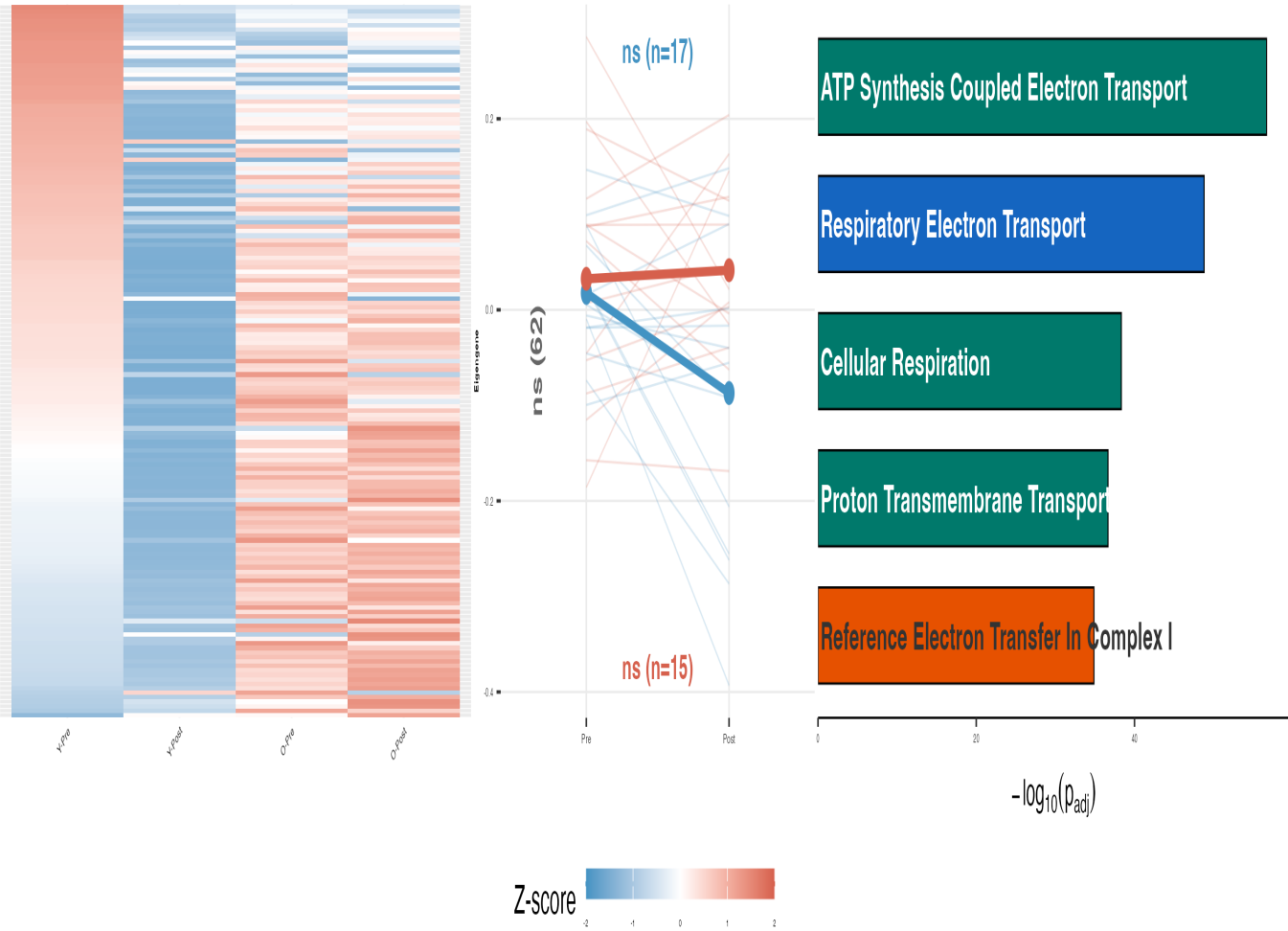
*FDR < 0.5 **FDR < 0.1 ***FDR < 0.01 (all LMM Bonferroni)

C

M5: Oxidative Phosphorylation

Training Young: $r = +0.45$

M5: Oxidative Phosphorylation (n=159)



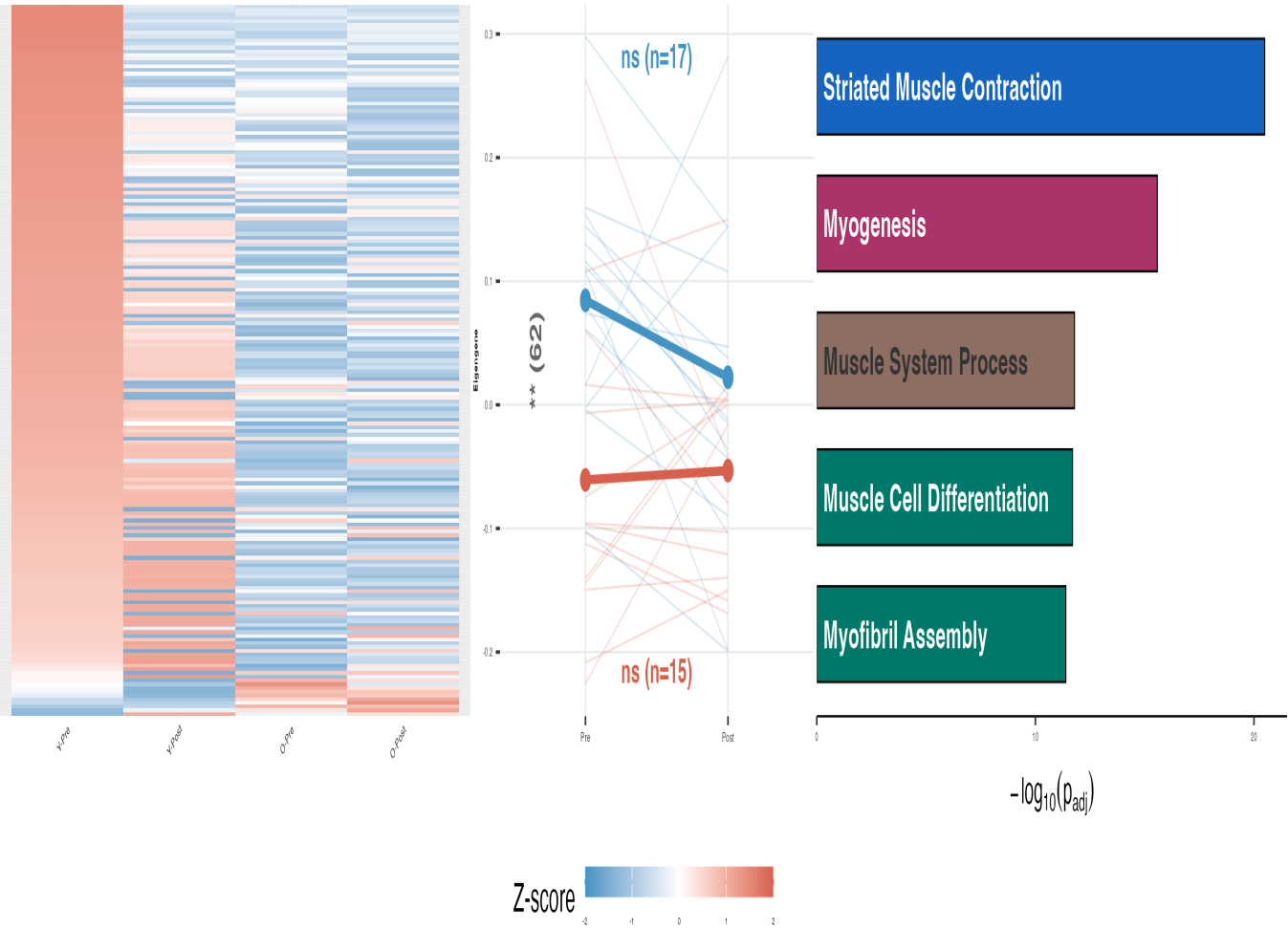
*FDR < 0.5 **FDR < 0.1 ***FDR < 0.01 (all LMM Bonferroni)

D

M3: Redox/Glycolysis

Aging: $r = -0.44$

M3: Redox/Glycolysis (n=191)



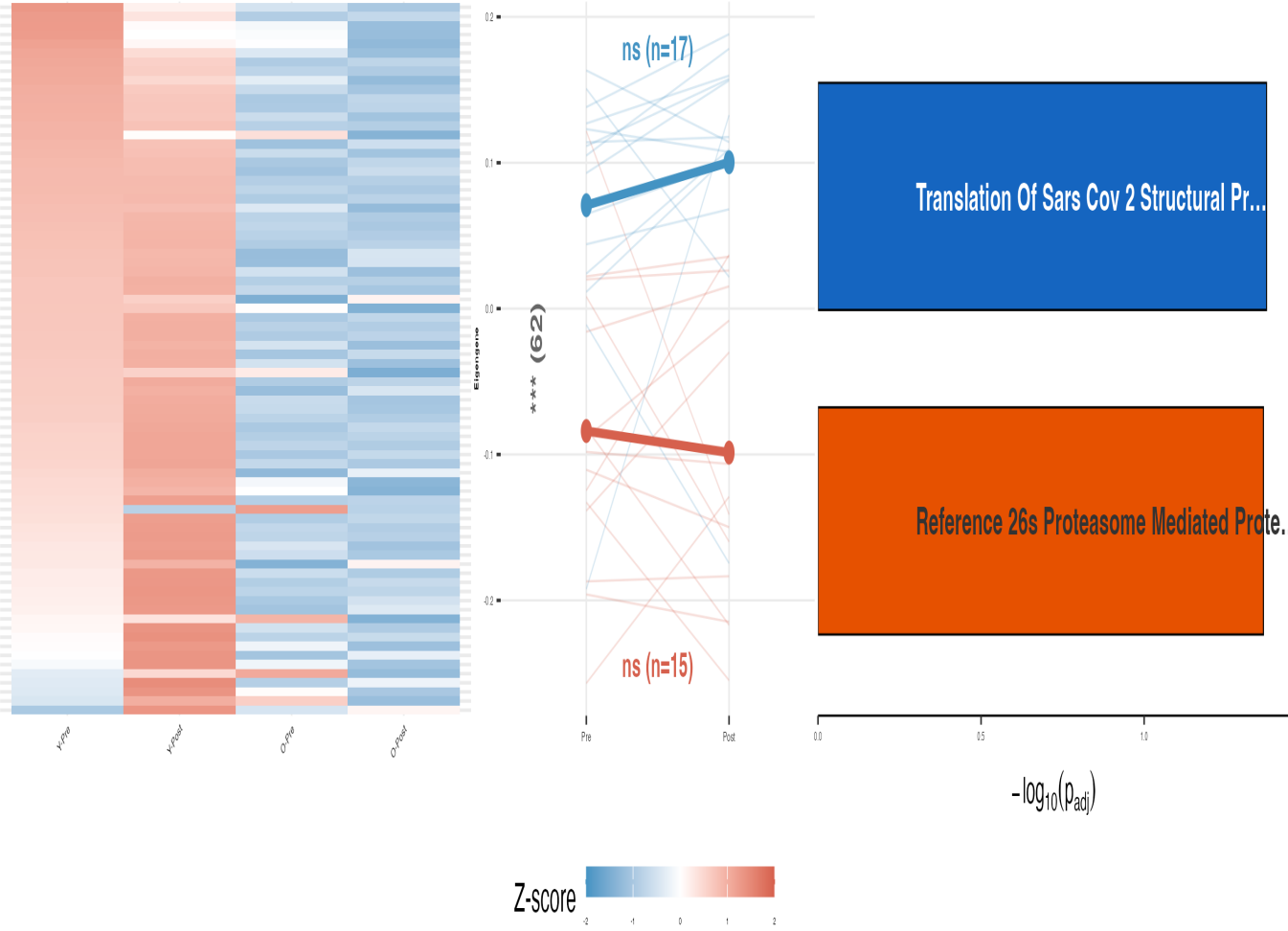
*FDR < 0.5 **FDR < 0.1 ***FDR < 0.01 (all LMM Bonferroni)

E

M9: Translation Initiation

Aging: $r = -0.53$

M9: Translation Initiation (n=78)

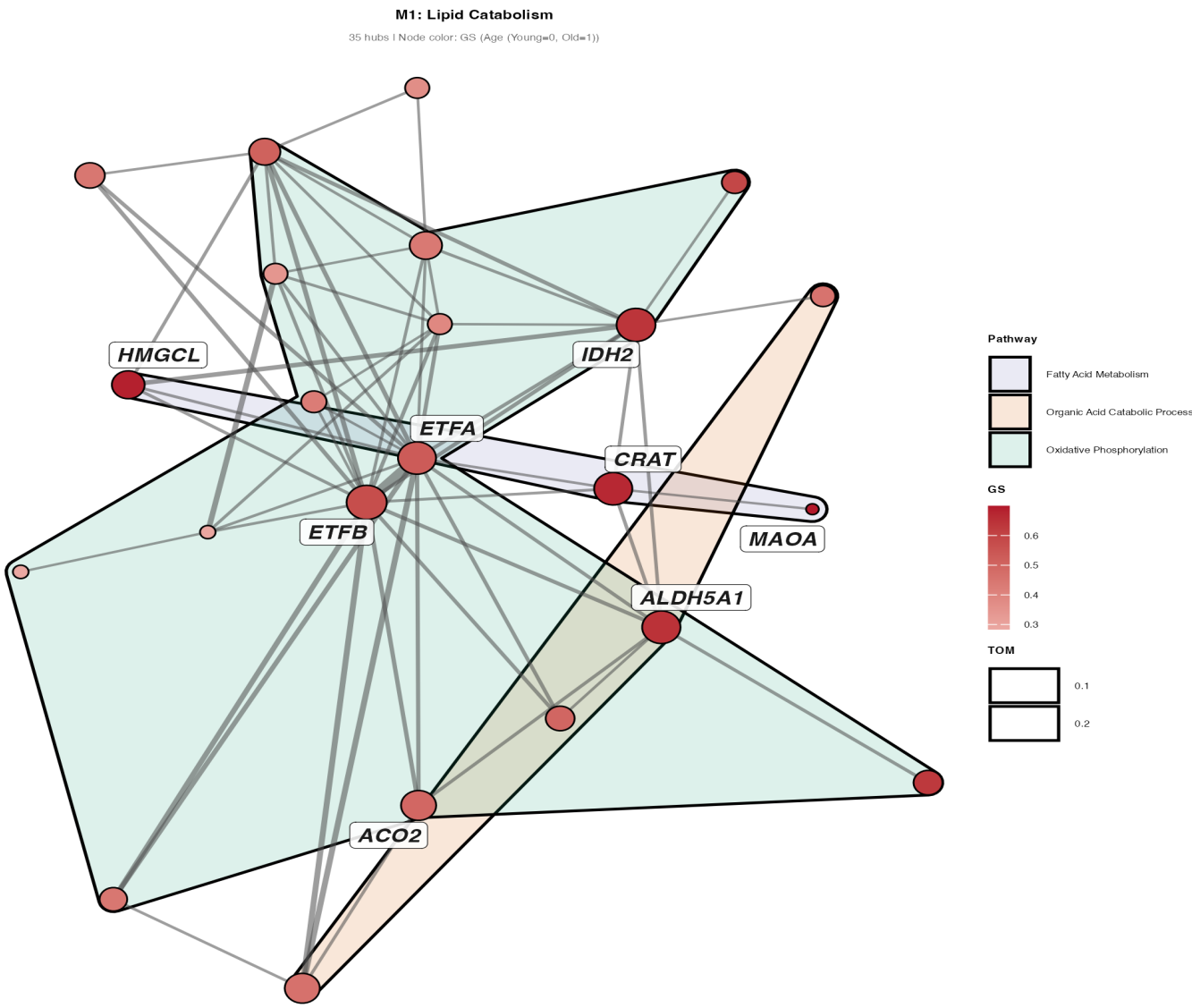


*FDR < 0.5 **FDR < 0.1 ***FDR < 0.01 (all LMM Bonferroni)

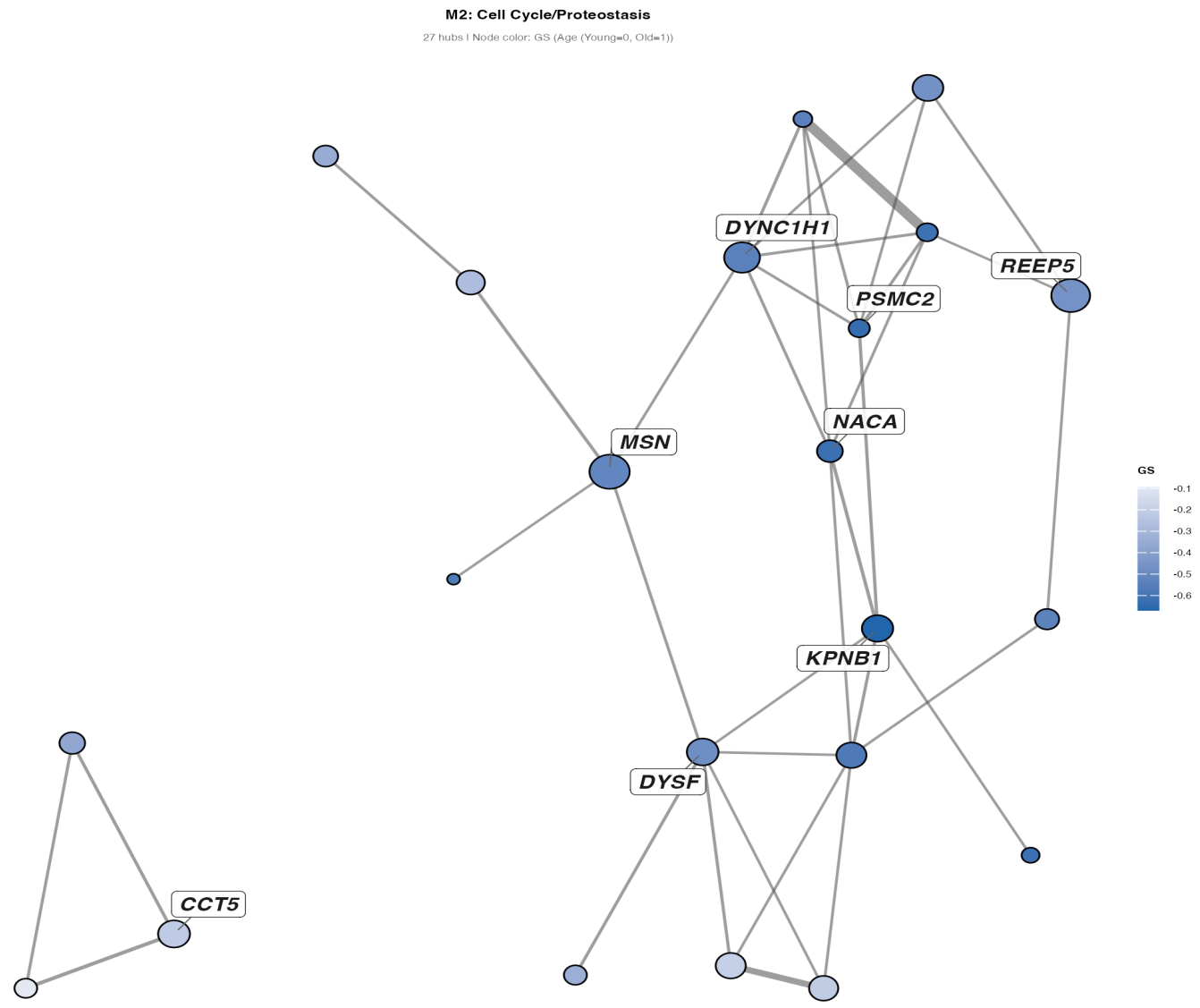
S5d – Hub Protein Networks (Key Modules)

A: Turquoise | B: Blue | C: Green | D: Brown | E: Magenta | F: Hub chord diagram

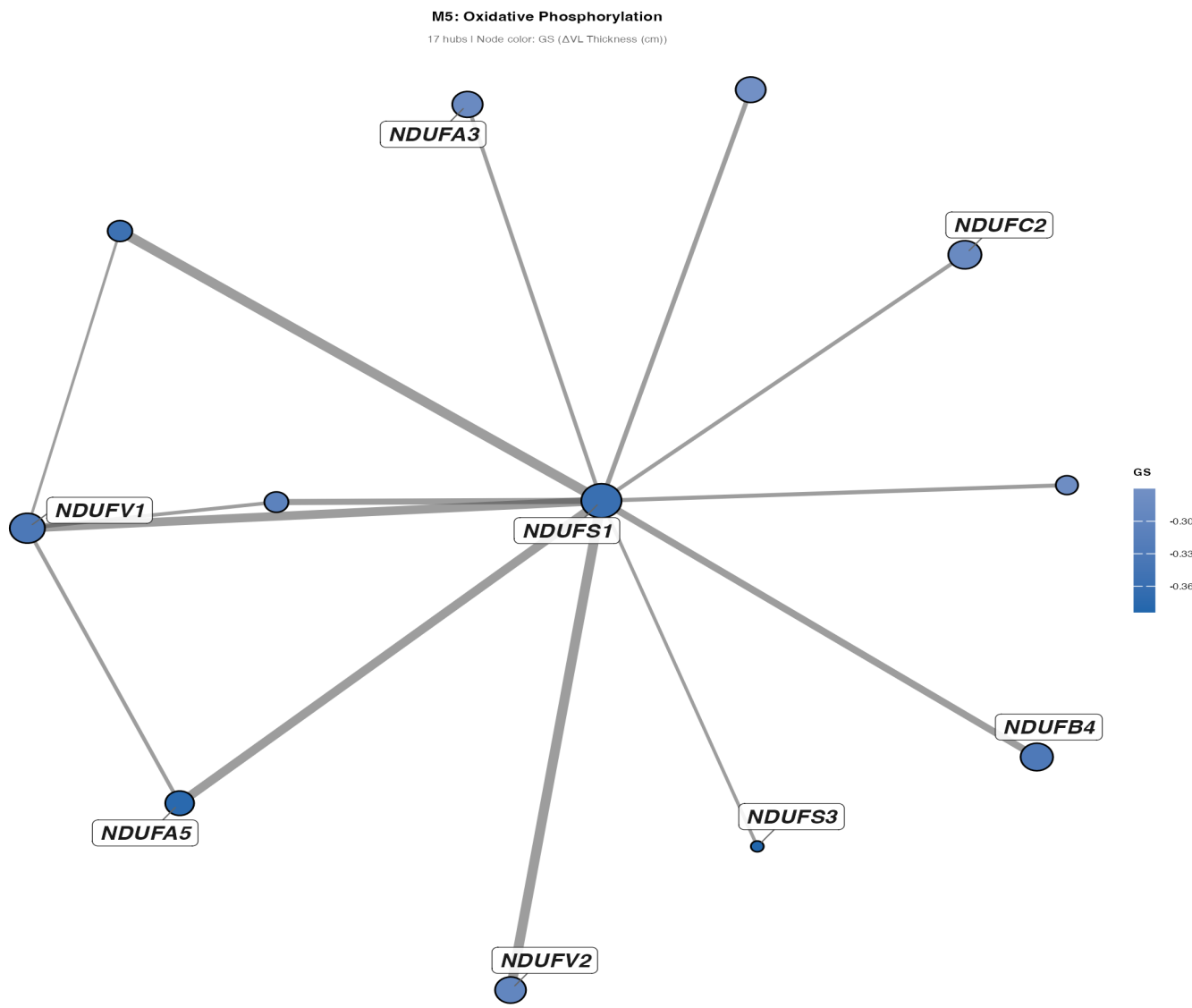
A



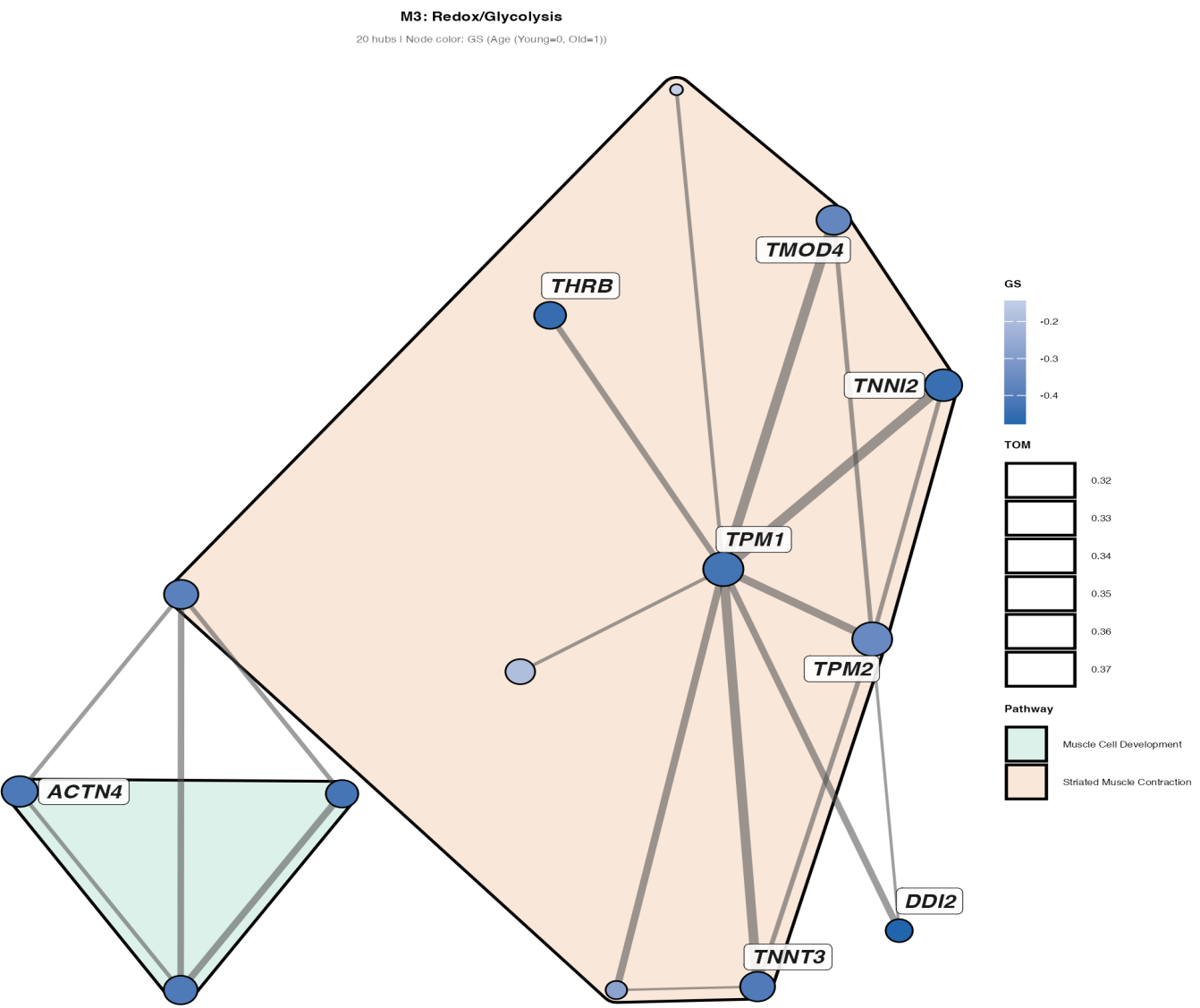
B



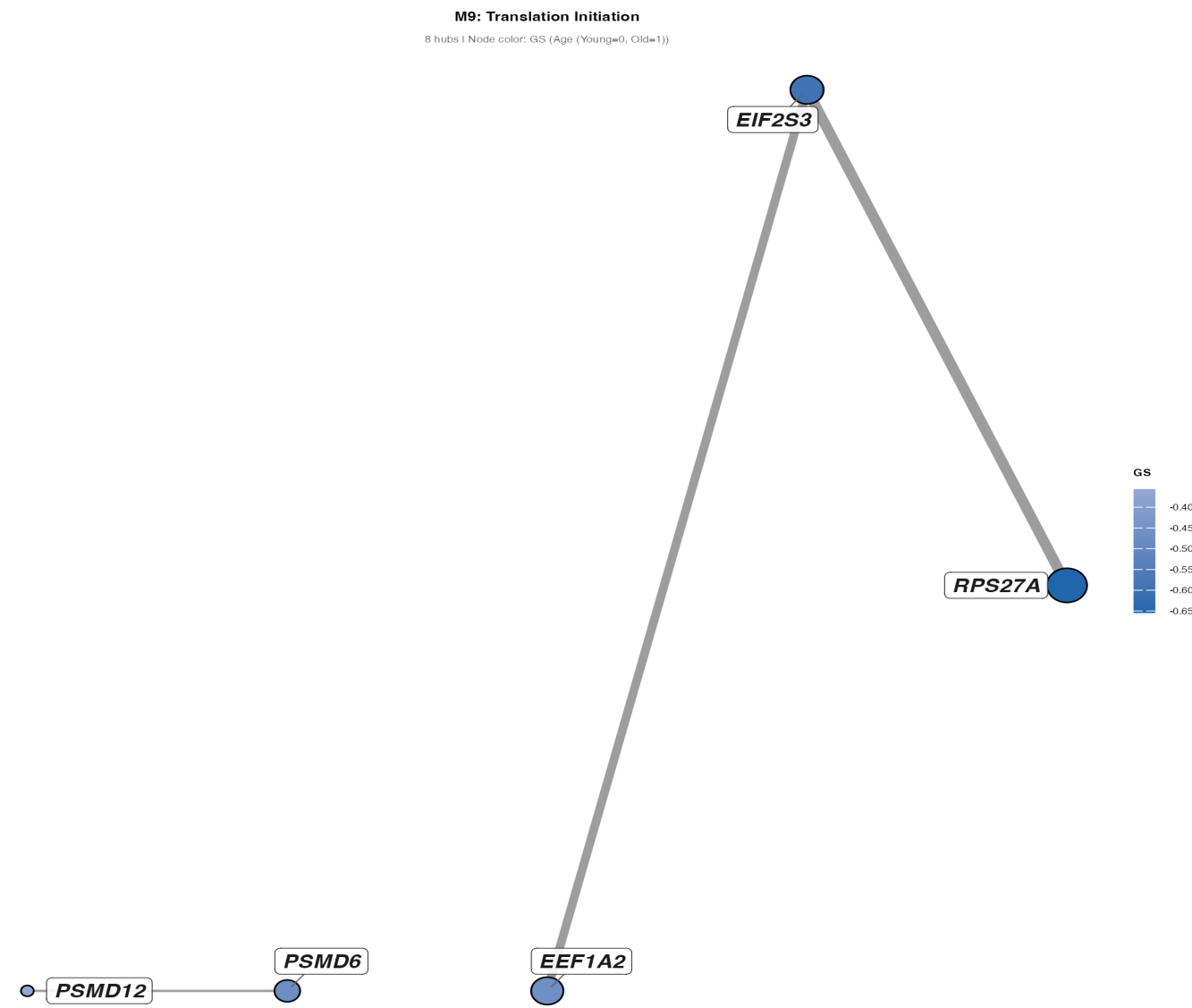
C



D



E



F

